Reliable Knowledge-Grounded AI Copilots for Dynamic Information Synthesis

Xinya Du, University of Texas at Dallas | ONR FY27 YIP Brief Topic Description

Prospective Program Officer: Dr. Rebecca Goolsby

Overview

Retrieval-augmented copilots promise rapid synthesis of heterogeneous reports, but retrieving relevant passages does not ensure that a generated conclusion is correct. A system may merge obsolete and current statements, confuse entities or events, overlook qualifications, or combine individually true facts into an unsupported claim. These failures worsen as information is updated.

Central research question: How can an AI copilot synthesize dynamic information, verify each substantive claim, and repair errors without introducing new unsupported conclusions?

We propose three connected research thrusts:

- **Structured integration of changing information.** We will extract entities, attributes, events, temporal relations, and evidence provenance from heterogeneous documents into updateable knowledge representations. The system will identify duplicated, conflicting, and superseded statements while retaining the context necessary for downstream reasoning.
- **Multi-step knowledge-grounded verification.** We will decompose synthesized outputs into checkable claims and determine which facts, relations, rules, and assumptions each requires. A verification planner will combine retrieval, knowledge-graph queries, and logical or numerical checks to classify claims as supported, contradicted, or unresolved.
- **Localized feedback and selective repair.** We will develop fine-grained feedback for entity, attribute, relation, factual, and logical consistency. Rather than regenerate the entire response, a copilot will seek missing evidence, revise affected claims, preserve supported content, and recheck downstream dependencies.

Evaluation. We will benchmark multi-report synthesis under stale evidence, contradictory updates, source duplication, and topic shifts. Metrics include claim support, logical consistency, robustness to updates, harmful corrections, and computation. Baselines include standard RAG, self-critique, iterative retrieval, and unstructured verification.

Naval relevance and potential impact. The research could support future information-analysis copilots for logistics, maintenance, and situation reports, where decision-makers must identify what is supported, what has changed, and which conclusions still require investigation.